Nebula Architecture Whitepaper

The architecture has two layers. The control plane schedules work and stores declarative configuration in etcd; three etcd nodes form a quorum.

The data plane executes inference and streams results. It is stateless except for a bounded request queue, so a data-plane node can be replaced at any time.

Every request carries a trace identifier that is propagated across services. Traces are sampled at 1 in 1000 by default.

A dedicated gateway terminates TLS and enforces per-tenant quotas before a request reaches the scheduler.